Patrick Thangarajah

Product Design | Prototyping | Testing & Validation | Simulation

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Hiring Team

11 November 2026

Planar Motor Inc.
Richmond, BC

Mechanical Design Engineer/EIT/Specialist

Dear Planar Motor Hiring Team,

I recently completed my MASc in Mechanical and Mechatronics Engineering at the University of Waterloo. I've always enjoyed the practical side of engineering, particularly designing and building hardware, then testing it to see how well it works. I was interested in Planar Motor's levitating movers because they combine mechanical design with motion control. I'm also interested in the opportunity to work on equipment used in different industries, including battery production and medical devices.

Through my experiences building test equipment, I have developed of my practical design. In my capstone project, I developed silicone and agar bladder phantoms and built the control and data acquisition interface for a pressurized test system. The completed system achieved stable, repeatable fill and void cycles up to 4.75 kPa. After the capstone, I continued work with the client, designing a PCB and writing five manufacturing and verification procedures for the electrical systems. Additionally, as a member of a UBC drone design team, I contributed to designing a payload drop system in SolidWorks and Onshape. By using several 3D-printed iterations and reusing components from previous designs, the team reduced its weight by 50%.

As a mechanical engineering intern at a manufacturing company, I wrote work instructions and SOPs using engineering drawings and customer specifications. More recently, my graduate research involved validating an LS-DYNA model against physical sled tests and investigating differences in restraint fit and response timing. I'm comfortable using SolidWorks, 3D printing, a lathe and a mill, and would like to gain more experience with CNC machining and welding.

I'm eligible to work in Canada and available to start immediately. I'd be happy to relocate to Richmond for this opportunity. I've attached my resume and transcripts and would appreciate the chance to discuss the position further.

Sincerely,

Patrick Thangarajah